

Computational Economics

An agent-based study on the dynamic distribution and firms concentration in a closed economy

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An agent-based study on the dynamic distribution and firms concentration in a closed economy

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Abstract

This paper studies an agent-based model for dynamic distribution of firms in a closed (supply) economy. Our model economy consists of N non-interacting firms. The Capital of each firm is driven by a multiplicative random walk with a drift. The initial capitals of the firms are homogeneously distributed on a closed interval. The dynamics is determined by two universal constants, the capital productivity and the expenditure coefficient, which are the same across the firms of the economy, and the price of the unit produced by the companies which undergoes stochastic fluctuations in a supply economy. We run Monte Carlo simulations in a finite time horizon and we find that the distribution of the final capitals of the firms not only it is not homogeneous but it may lead to monopolies. The relation between the universal constants and the stochastic process **determines the phases of the business cycle undergone by the firms of the economy** as well as the final distribution of the capitals of the firms in the economy. We use various metrics to relate the final distribution of the capitals with the capital productivity, the expenditure coefficient and the probability distribution of the unit price.

JEL Classification: C63; D10

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1 Introduction

New methods of economic modelling have been sought as a result of the global economic downturn in 2008. ^{spelling} Hamill and Gilbert (2015) highlight the benefits of agent-based models and demonstrate how these can easily handle complexity with heterogeneous agents, households, and firms that interact dynamically. Unlike traditional methods, agent-based models (Axtell and Farmer (2022), Buchanan (2009), Farmer and Foley (2009), Hamil and Gilbert (2015), Holcombe et al (2013), Macal and North (2005), Tesfatsion and Judd (2006)) do not require people or firms to optimize or economic systems to reach equilibrium. Furthermore, they offer a way to link micro foundations directly to the macro-level. As such agent-based models have found applications also in operation research (McGroarty et al (2018), ^{year mismatch} Rand and Stummer (2021), Bunn and Oliveira (2003)).

A salient feature of agent-based models is that the aggregate level is not explicitly pre-specified but rather it *emerges* from the myriad of interactions between the agents. This multiplicity of interactions gives rise to a large amount of complexity that can be manifested in many forms including path-dependence, difficulty of prediction, and qualities that are not analytically tractable from system components and their attributes alone.

In this paper we use an agent based model that aims to capture the dynamics of a market with a fixed number of firms. The model includes a set of parameters, such as the number of firms, the range of capital levels, and the coefficients that determine the impact of prices and costs on production decisions. These parameters can be adjusted to explore how different market structures and conditions affect firm behavior and market outcomes.

One key aspect of the simulation model is its use of the Hirschman-Herfindahl Index (HHI) to measure market concentration. The HHI is a commonly used measure of market concentration that takes into account the market share of each firm in the market. The HHI ranges from 0 to 1, with higher values indicating greater market concentration. The use of the HHI is consistent with industrial organization theory, which emphasizes the importance of market structure and its impact on firm behavior and market outcomes. Here we have used the modified Hirschman-Herfindahl Index (MHHI) is a variation of the HHI that incorporates weights to adjust for differences in the size of firms in a market. The formula for the MHHI is:

$$MHHI = \sum (S_i w_i)^2$$

where S_i is the market share of firm i , and w_i is a weight that reflects the size of firm i relative to the overall size of the market.

Weights can be calculated in different ways depending on the specific context of the market being studied. For example, one common method is to use the ratio

of the size of each firm to the total size of the market as the weight. Another method is to use the inverse of the number of firms in the market as the weight, which gives more weight to larger firms.

The use of weights in the MHHI allows for a more nuanced analysis of market concentration, as it takes into account the size of firms in addition to their market share. This can be particularly important in markets with large disparities in firm size, where a simple market share-based measure such as the HHI may not accurately capture the true level of concentration.

Another important feature of the simulation model is its incorporation of risk, which is calculated as the standard deviation of the prices that the firms charge for their products. Higher levels of risk can indicate greater volatility in the market, which can affect firm behavior and market outcomes. This aspect of the simulation model is consistent with behavioral economics, which recognizes that decision-making is influenced by factors such as risk aversion and bounded rationality. need reference

The simulation model also reflects the neoclassical theory of the firm, which assumes that firms seek to maximize profits by choosing the level of output that maximizes the difference between revenue and costs. The model incorporates the impact of prices and costs on production decisions, which is consistent with the neoclassical view of the firm as a profit-maximizing entity. However, the model also recognizes that firm behavior is influenced by market structure and risk, which are important considerations in real-world markets. check the result

Overall, the simulation model provides a useful tool for studying the dynamics of a market with a fixed number of firms and can help researchers better understand the factors that drive firm behavior and market outcomes. The incorporation of economic theory, such as industrial organization theory, behavioral economics, and neoclassical theory, provides a strong theoretical foundation for the simulation model and enhances its relevance to real-world markets.

Several studies have examined market concentration and its implications for economic outcomes. For example, Scherer and Ross (1990) highlight the relationship between market concentration and market power, suggesting that highly concentrated markets may be characterized by lower levels of competition and higher prices. Similarly, (Aghion et al. (2005)) double parentheses find that market concentration may affect innovation, with highly concentrated markets potentially discouraging innovation due to reduced competitive pressures.

The paper is organized as follows. Section 2 describes the general setup of the model. Section 3 introduces the simulations and the firm dynamics. Section 4 gives the dynamics and the terminal distribution of capitals. Section 5 gives the terminal distribution of companies by convolution. Section 6 gives the frequency distribution of the HMMI index are not introduced in the presence of sectors. Section 7 concludes and outlines prospects for future research.

2 The general setup

Consider an economy consisting of N firms indexed by $i = 1, 2, \dots, N$. The Capital of the i^{th} firm at time $t + 1$ is given by

$$C_{i,t+1} = C_{i,t} + (p_{i,t}\phi - c)C_{i,t}, \quad (2.1)$$

where ϕ is a capital productivity coefficient, c is an expenditure coefficient, and $p_{i,t}$ is the price of the unit produced by the i^{th} firm at time t . Equation (2.1) is a simplified representation of a capital production function that calculates a firm's new level of capital $C_{i,t+1}$ based on its current level of capital $C_{i,t}$, the price $p_{i,t}$ it charges for its product, and its costs. Specifically, the formula calculates the firm's new level of capital $C_{i,t+1}$ by adding to its current capital $C_{i,t}$ the difference between its revenue (which is equal to the price it charges for its product multiplied by its level of output) and its costs (which are represented by the term $cC_{i,t}$ in the formula).

We assume that the price $p_{i,t}$ undergoes stochastic fluctuations and that the firms are non-interacting. Moreover, We assume that

check theory for this assumption, is it reflect the market? $\bar{p}\phi - c > 0, \quad (2.2)$

where $\bar{p}$ is the mean value of p .

From equation (2.1) we infer that

$$C_{i,t+1} = (1 + d_{i,t})C_{i,t}, \quad (2.3)$$

where $d_{i,t} = p_{i,t}\phi - c$. Therefore the capital undergoes multiplicative random walk with a drift. The initial capitals of the firms are homogeneously distributed in the interval $[1000, 6000]$. The dynamics of our economy is determined from the combination $d_{i,t} = p_{i,t}\phi - c$. If $d_{i,t}$ were a constant d the difference equation (2.3) would give the trend

$$C_{i,t+1} = C_{i,0}(1 + d)^t,$$

of the evolution of the capital of the i^{th} firm as a function of time.

By taking the logarithm of equation (2.3) we obtain

$$\log C_{i,t+1} = \log(1 + d_{i,t}) + \log C_{i,t}. \quad (2.4)$$

The combination $d_{i,t} = p_{i,t}\phi - c$ determines the four major phases of the business cycles of the firms of the economy as well as the distribution of the capitals of the firms at the end of a finite time horizon.

3 Simulations and Firm Dynamics

We run Monte Carlo simulations for $N = 100$ firms with initial capitals homogeneously distributed in the interval $[1000, 5000]$. We depict equation (2.4) for the sum of the capitals of the firms of the economy. Every simulation is characterised by a vector $v = (v_1, v_2, v_3, v_4, v_5, v_6)$ with six components. The meaning of these components is as follows:

1. v_1 is the number of firms in the final distribution with capital greater than the 40% of the mean of the capitals of the firms.
2. v_2 is the modified Herfindahl-Hirschman Index.
3. v_3 is the logarithm of the mean of the capitals of the firms in the final distribution of the capitals of the firms.
4. v_4 is the standard deviation of the price of the unit which undergoes stochastic fluctuations.
5. $v_5 = \phi$ is the capital productivity coefficient.
6. $v_6 = c$ is the expenditure coefficient.

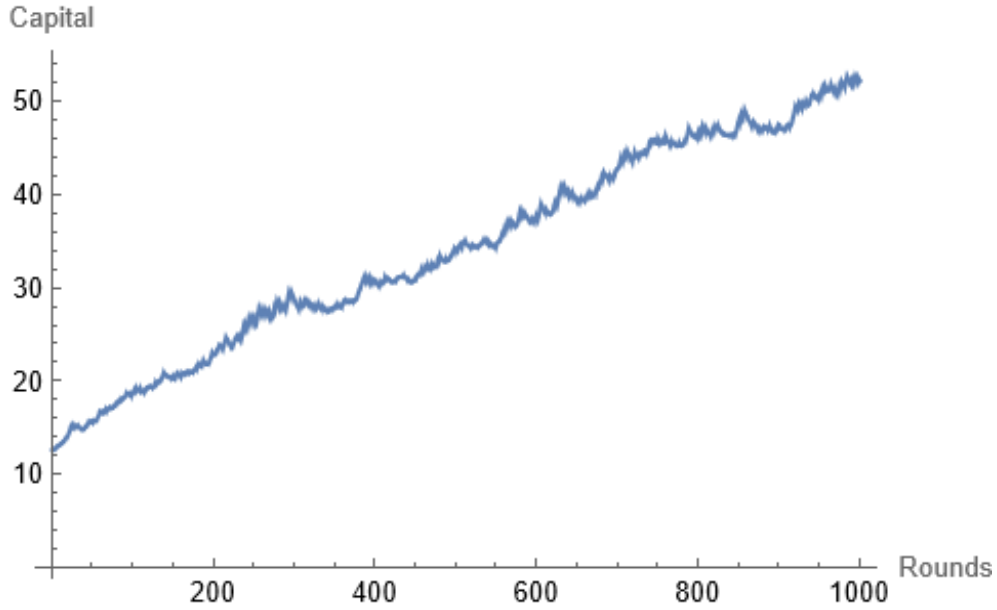


Figure 1: $(v_1, v_2, v_3, v_4, v_5, v_6) = (8, 0.702268, 47.4516, 0.895031, 0.4, 0.2)$.

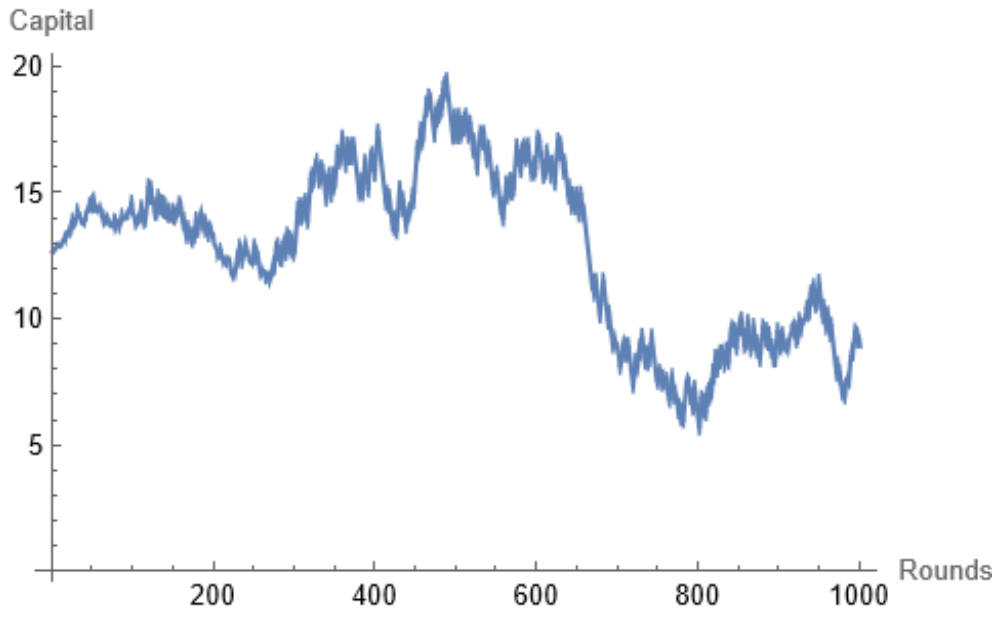


Figure 2: $(v_1, v_2, v_3, v_4, v_5, v_6) = (2, 0.622217, 4.30743, 0.887447, 0.4, 0.25)$.

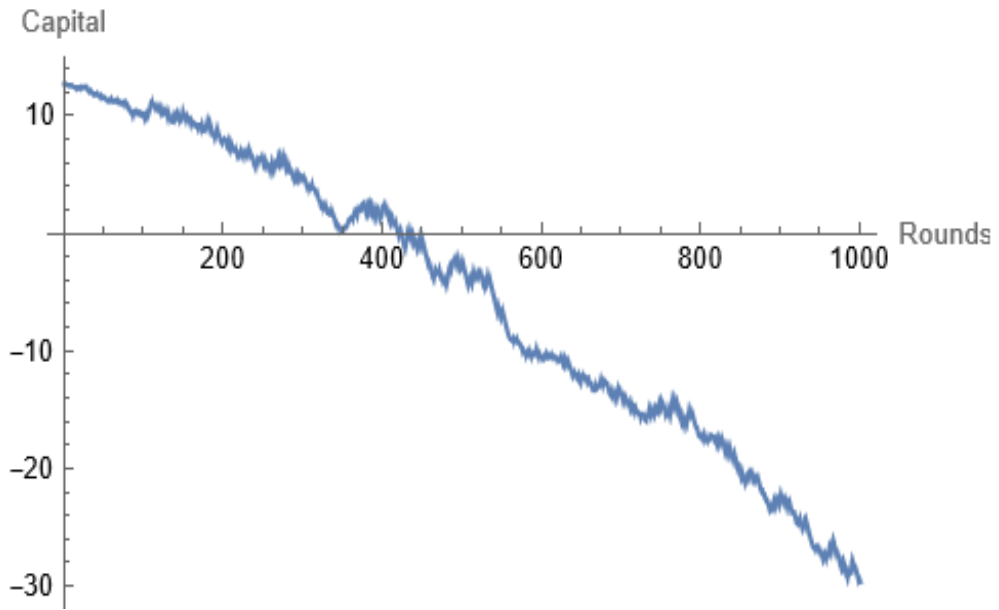


Figure 3: $(v_1, v_2, v_3, v_4, v_5, v_6) = (6, 0.779583, -34.2565, 0.792636, 0.4, 0.25)$.

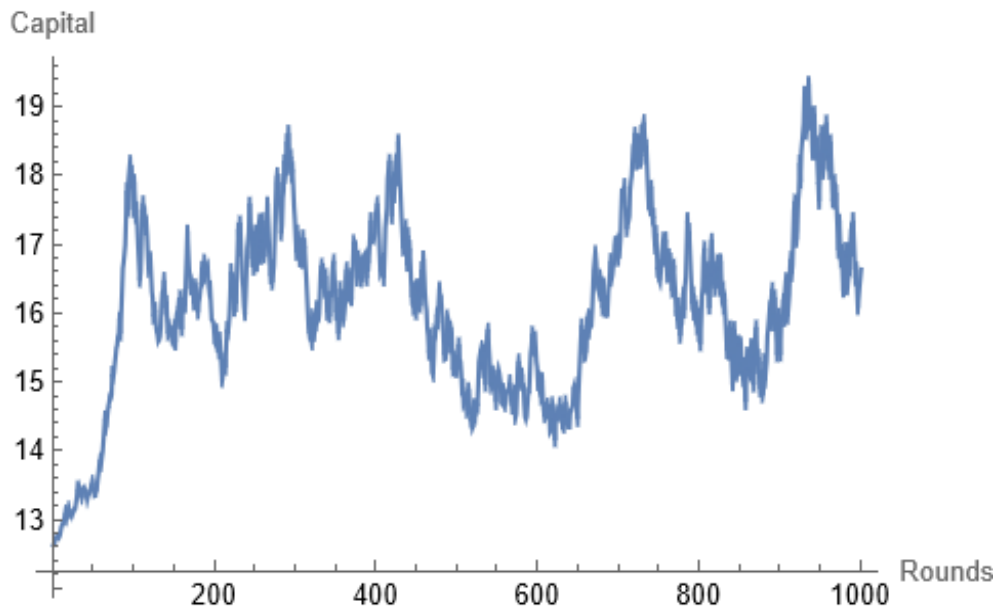


Figure 4: $(v_1, v_2, v_3, v_4, v_5, v_6) = (2, 0.982597, 12.0254, 0.766824, 0.4, 0.25)$.

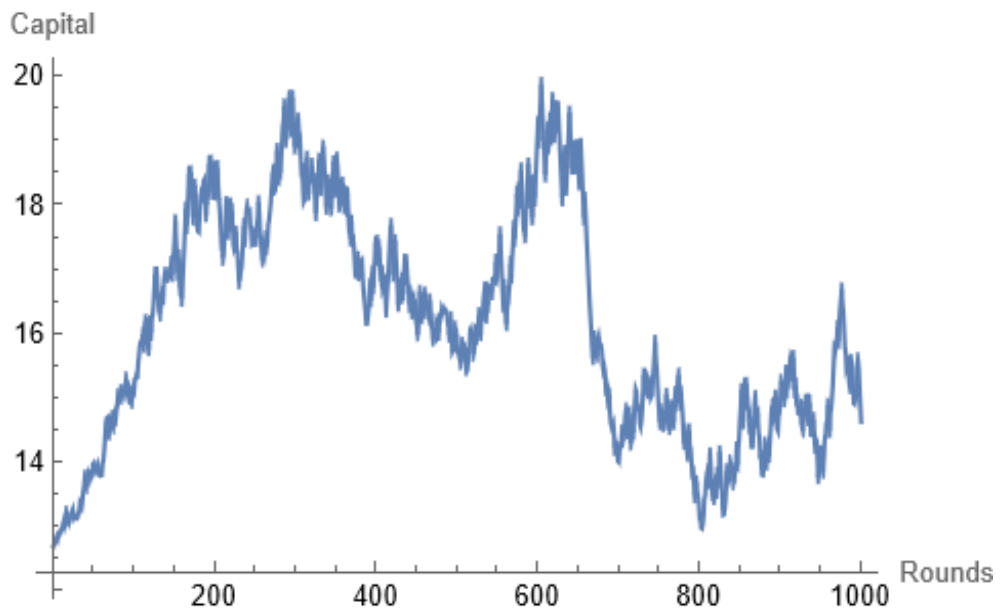


Figure 5: $(v_1, v_2, v_3, v_4, v_5, v_6) = (3, 0.652818, 10.0041, 0.712196, 0.4, 0.25)$.

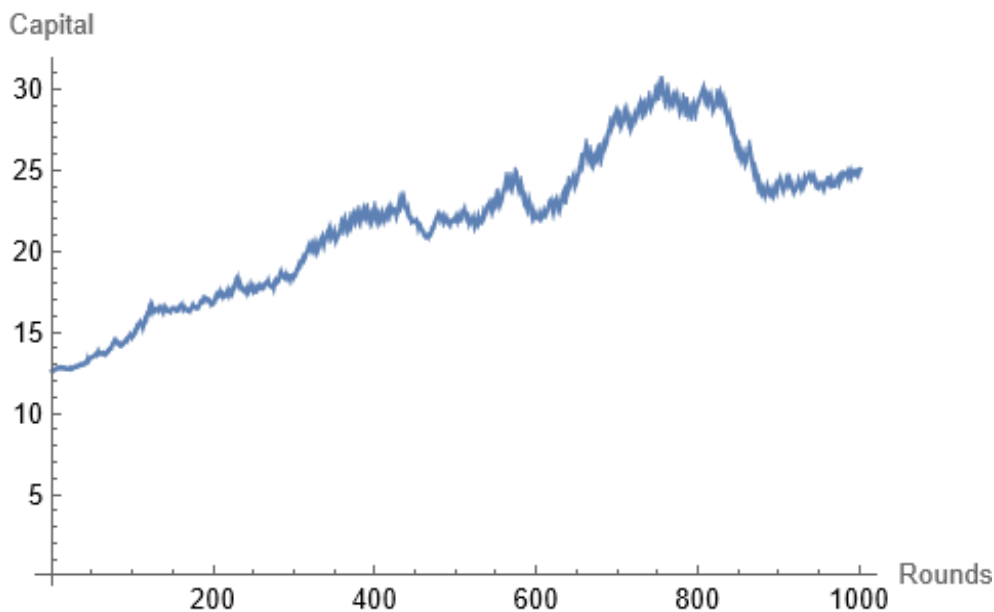


Figure 6: $(v_1, v_2, v_3, v_4, v_5, v_6) = (6, 0.634695, 20.4516, 0.602278, 0.4, 0.25)$.

4 Dynamics and terminal distribution of capitals

In order to better understand the dynamics of our model we study further the Firm dynamics in a set of 300 companies executing 1000 time steps for 500 rounds in Monte Carlo simulation. We relate characteristics of the final distribution of the companies in their accumulated capital with structural properties of the model. In particular we perform regression analysis of two characteristics of the final distribution of the companies, namely the number of companies whose final capital is greater than the 40% of the mean capital in the final distribution of the firms of our economy and the MHHI index in the final distribution of the firms of our economy, using as **explanatory variables** the two universal constants ϕ and c , the logarithm of the initial total capital of our economy and the standard deviation of the distribution of the **price $p_{i,t}$** at time t of the unit produced by the i^{th} firm. Moreover, we study the distribution of the MHHI index in the final distribution of the firms of our economy after executing 1000 time steps for 500 rounds in Monte Carlo simulation. We obtain different distributions for the MHHI index for different values of ϕ and c .

4.1 Regression analysis

The Correlation matrix of the six variables $v_1, v_2, v_3, v_4, v_5, v_6$ defined in Section

3 is given below.

$$\begin{pmatrix} 1. & -0.632364 & -0.18791 & -0.224881 & -0.284757 & -0.105886 \\ -0.632364 & 1. & 0.142358 & 0.0935905 & 0.125636 & -0.0245894 \\ -0.18791 & 0.142358 & 1. & 0.582909 & 0.360748 & -0.686991 \\ -0.224881 & 0.0935905 & 0.582909 & 1. & 0.0146498 & 0.0330434 \\ -0.284757 & 0.125636 & 0.360748 & 0.0146498 & 1. & -0.0238712 \\ -0.105886 & -0.0245894 & -0.686991 & 0.0330434 & -0.0238712 & 1. \end{pmatrix}$$

Figure 7: Matrix correlation of the variables $v_1, v_2, v_3, v_4, v_5, v_6$.

4.1.1 Regression analysis of the number of companies which are greater than 0.4 of the mean of capital index

We use regression analysis in order to interpret the number of companies whose final capital is greater than the 40% of the mean capital in the final distribution of the firms of our economy (variable v_1) in terms of the variables v_3, v_4, v_5, v_6 defined in Section 3. The statistics of the regression is given in the following table.

	Estimate	Standard Error	t-Statistic	P-Value
#1	-0.0615415	0.0049898	-12.3335	1.13715×10^{-30}
#2	17.1085	1.1311	15.1255	8.92552×10^{-43}
#3	1.99304	0.981475	2.03066	0.0428229
#4	-35.4122	3.33266	-10.6258	6.71097×10^{-24}

Figure 8: Statistics of the regression of v_1 in terms of v_3, v_4, v_5, v_6 .

The linear model which interpolates the data is

$$v_1 = c_3 v_3 + c_4 v_4 + c_5 v_5 + c_6 v_6. \quad (4.1)$$

As depicted in Figure 8 $(c_3, c_4, c_5, c_6) = (-0.0615415, 17.1085, 1.99304, -35.4122)$. The adjusted R-squared is 0.701756. After normalization we have $(c_3, c_4, c_5, c_6) = (-3.4642, 14.2715, 1.40099, -6.24083)$.

do you mean v3?

Therefore variables v_4 and v_5 have positive relation with v_1 , whereas v_3 and v_6 have negative relation with v_1 with the most important variable being v_4 which is the standard deviation of price.

4.1.2 Regression analysis of the HHIM index

We use regression analysis in order to interpret the HHIM index in the final distribution of the firms of our economy (variable v_2) in terms of the variables v_3, v_4, v_5, v_6 defined in Section 3. The statistics of the regression is given in the following table.

	Estimate	Standard Error	t-Statistic	P-Value
#1	0.0113206	0.00289384	3.91198	0.000170888
#2	-0.0257125	0.246112	-0.104475	0.917011
#3	-2.94661	0.779203	-3.78157	0.000270849
#4	4.74502	1.12869	4.204	0.0000588494

Figure 9: Statistics of the regression of v_2 in terms of v_3, v_4, v_5, v_6 .

The linear model which interpolates the data is

$$v_2 = c_3 v_3 + c_4 v_4 + c_5 v_5 + c_6 v_6. \quad (4.2)$$

As depicted in Figure 9, $(c_3, c_4, c_5, c_6) = (-0.00141533, 0.882429, 0.39555, -0.72071)$. The adjusted R-squared is 0.949392. After normalization we have $(c_3, c_4, c_5, c_6) = (-0.0796695, 0.7361, 0.278049, -0.127013)$. From the statistics of Figure 9 we infer that the variable v_4 is not statistically important. So we do again regression analysis omitting this variable. The Statistics for the new modified regression is given by

	Estimate	Standard Error	t-Statistic	P-Value
#1	0.00108183	0.00022137	4.88695	1.38392×10^{-6}
#2	0.750655	0.0502868	14.9275	6.74858×10^{-42}
#3	1.07653	0.13927	7.72983	6.01424×10^{-14}

Figure 10: Statistics of the regression of v_2 in terms of v_3, v_5, v_6 .

The linear model which interpolates the data is

$$v_2 = c_3 v_3 + c_5 v_5 + c_6 v_6. \quad (4.3)$$

After normalization we find $(c_3, c_5, c_6) = (0.0608965, 0.626177, 0.756741)$. The adjusted R-squared is 0.920316.

4.2 Simulated Distribution of the HMMI index

We give now the distribution of the MHHI index for 300 companies and 1000 time steps and 500 rounds of the Monte Carlo simulation for various values of the of the capital productivity coefficient ϕ and the expenditure coefficient c .

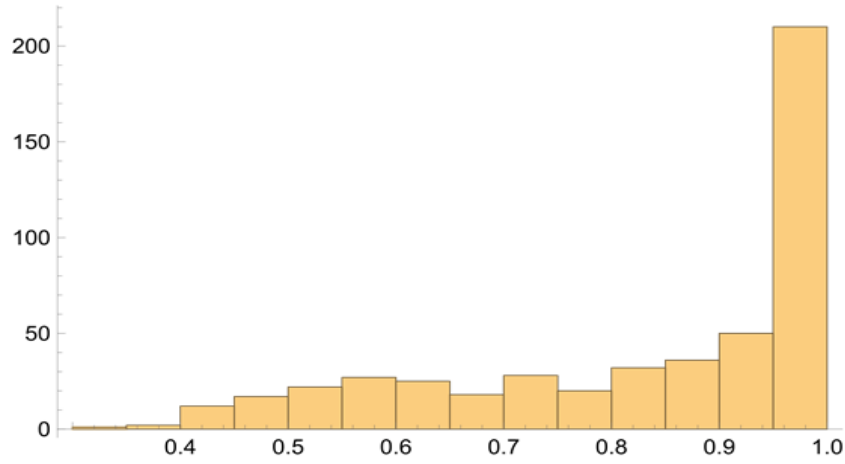


Figure 11: Distribution of the MHHI index for $\phi = 0.81, c = 0.28$.

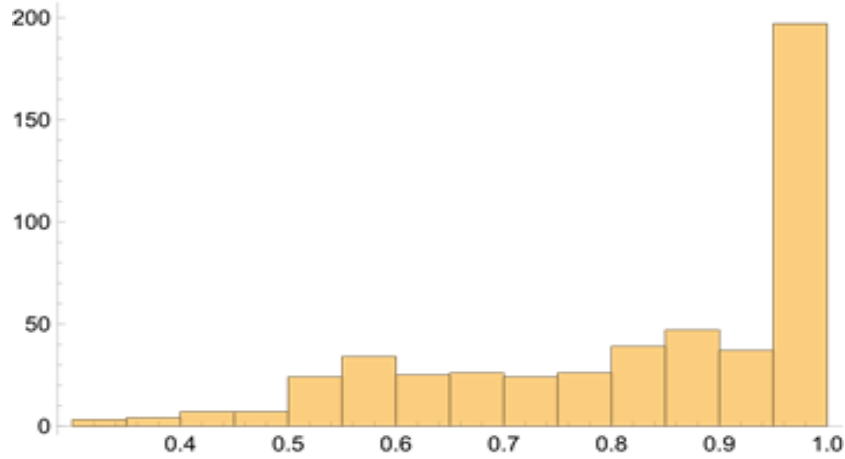


Figure 12: Distribution of the MHHI index for $\phi = 0.42, c = 0.35$.

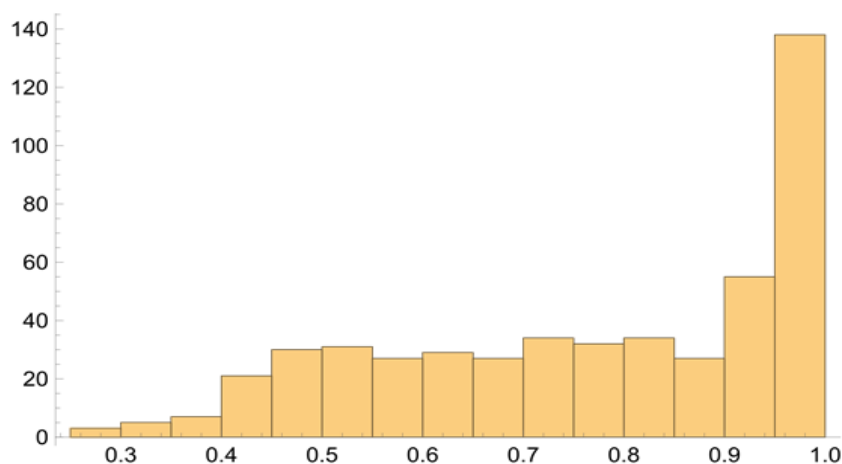


Figure 13: Distribution of the MHHI index for $\phi = 0.72, c = 0.14$.

As depicted in Figures 11, 12 and 13, a mechanical result of multiplicative growth for different values of ϕ and c , in the majority of cases, the Economy is driven towards oligopoly. The companies of the oligopoly which have the greatest share in the final market concentration are precisely those with the larger value of $d_{i,t} = p_{i,t}\phi - c$ in most of the 1000 time steps.

5 The terminal distribution of companies by convolution

The simple multiplicative structure of this model suggests that the terminal pdf of the capitals of companies that evolve according to this model may be calculated by convolving the pdf of the initial capitals and that of the terminal value of a normalised random walk. We start with companies whose capital C_0 is a random variable which follows the uniform distribution $U(1000, 5000)$. We want to calculate the distribution of capital after 1000 time steps. The capital is given as the product of two random variables $C_{1000} = C_0 MRW_{1000}$, where MRW is Multiplicative Random Walk with initial value 1. Moreover, the terminal value $\log MRW_{1000}$ follows normal distribution $N(45, 11.5)$.

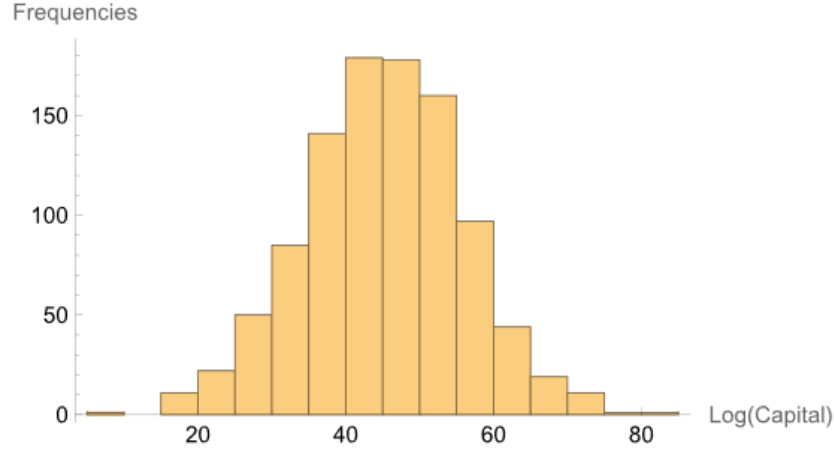


Figure 14: Histogram of log Capital

The distribution of C_{1000} is the product of two independent random variables C_0 and MRW_{1000} . For convenience we take

$$\log C_{1000} = \log C_0 + \log MRW_{1000}.$$

The pdf of $\log C_{1000}$ is given by the following integral

$$f(z) = \frac{1}{4000} \frac{1}{\sigma\sqrt{2\pi}} \cdot \int_{1000}^{6000} e^{-\frac{(z-x-\mu)^2}{2\sigma^2} + x} dx. \quad (5.1)$$

The pdf in Equation (5.1) is displayed in Figure 15. We can then use standard methods to estimate the pdf of the MHHI dispersion index.

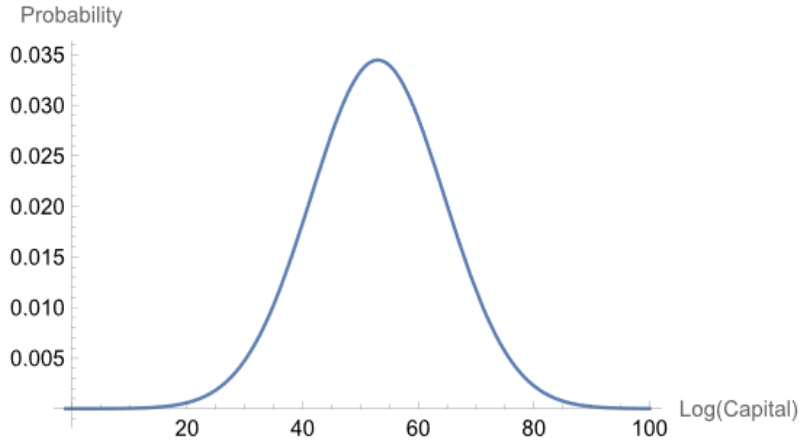


Figure 15: Pdf of the MHHI index.

6 Frequency distribution of the HMMI index in the presence of sectors

In order to better elucidate the dynamics of our model we partition 300 companies into 10 sectors and consider that the values of ϕ and c are different for each sector. We run the Monte Carlo simulation for 1000 time steps 500 times. We so obtain an absolute frequency distribution of the HMMI index, depicted below, which provides information for the market concentration in the final state of our economy.

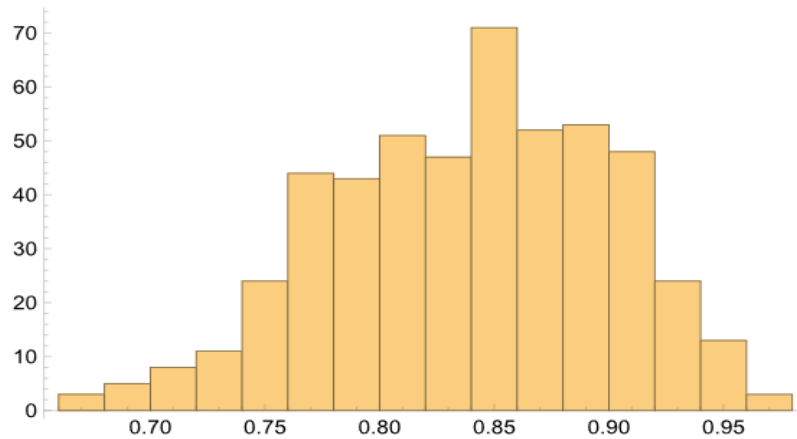


Figure 16: Absolute frequency distribution of HMMI index in the presence of 10 sectors in the Economy.

The best probability density function approximation to the distribution in Figure 16 is a Weibull(11.7,0.91) distribution displayed in the Figure which follows.

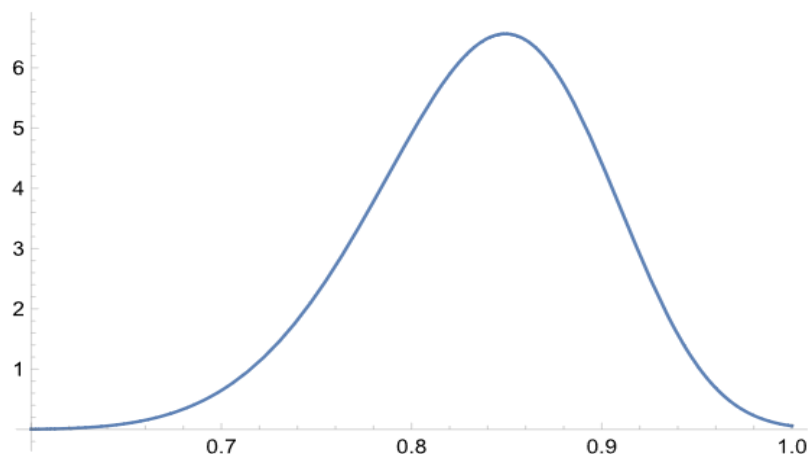


Figure 17: Probability density function of the HMMI index of the 300 companies of the Economy in the presence of 10 sectors.

Figure 16 is to be juxtaposed with Figures 11, 12 and 13. The main conclusion to be drawn is that as expected **the presence of sectors eradicates the emergence of big oligopolies or monopolies in the dynamics of our model postulated in Equation (1).**

this is not an emergent mechanism, but a mechanical averaging effect

Our results have several implications for economic theory. First, the relationship between price and cost coefficients and market concentration supports the idea that market power plays a crucial role in determining market structure (Stigler, 1950). Firms with higher profit margins, driven by favorable price and cost conditions, are more likely to dominate the market and contribute to a higher concentration. Second, the distribution of the MHHI can provide insights into the level of competition within an industry.

A higher MHHI indicates a more concentrated market, which may be associated with less competition, lower efficiency, and potentially higher prices for consumers (Boone, 2008). Third, the simulation results can inform policymakers about the potential impact of changes in market parameters on market concentration. By understanding the factors that drive market concentration, policymakers can design regulations and interventions that promote competition and market efficiency (Baker, 2019).

Our simulation results show that the distribution of the MHHI varies depending on the parameters used in the model. In general, the MHHI index tends to increase along with the increase of ϕ and the decrease of c . This is consistent with economic theory, which suggests that firms with higher profit margins tend to dominate the market and contribute to a higher concentration (Sutton, 1991).

7 Conclusion

We create business cycles in a model economy of 100 firms not via the interaction of the firms but instead via a multiplicative random walk of the capital of the firms which is induced by the stochastic fluctuations of the price of the unit produced by the firms. We run Monte Carlo simulations and generate the final state of the economy in a finite time horizon as an aggregate of the random walks of the capitals of the firms. The dynamics in our model economy is determined by the interplay of two constants, the capital productivity coefficient ϕ the expenditure coefficient c , and the probability distribution of the price $p_{i,t}$ of the unit produced by the firms. We relate characteristics of the final distribution of the capitals of the firms with the two constants and a few moments of the probability distribution of the price of the unit produced by the firms. As expected the emerging business cycles are determined by the relative values of ϕ , c , and $p_{i,t}$. For given ϕ and c large values of the standard deviation in the probability distribution of $p_{i,t}$ lead to downward trend in the emerging business cycle whereas small values of the standard deviation in the probability distribution of $p_{i,t}$ give rise to upward trend in the emerging business cycle.

We elucidate further the dynamics of our model by considering 300 firms running the Monte Carlo simulation for 1000 time steps 500 times. We study the final market concentration by constructing absolute frequency distributions of the MHHI index for various values of ϕ and c . Each time the values of ϕ and c are assumed to be the same for all the firms. As it is depicted in Figures 11, 12 and 13 in the majority of cases we are led to oligopolies where the market is shared between only a few companies for different values of ϕ and c . These are the companies with the most favourable values of $d_{i,t} = p_{i,t}\phi - c$, i.e., with large positive values of $d_{i,t}$ in most of the 1000 time steps.

The situation, as far as the formation of oligopolies is concerned, changes when we consider different sectors in our economy, where each sector comprises of 10 companies and the values of ϕ and c change from sector to sector. Thus we have 30 different sectors with 30 different values of pairs (ϕ, c) , one pair for each sector. Then the absolute frequency distribution of the HMMI index displayed in Figure 16 depicts a final state of the economy better than the formation of large oligopolies or monopolies. As expected the different values of ϕ and c for each sector smears out big oligopolies and monopolies and smoothens singular market concentration into more uniform and homogeneous market concentration.

The simulation findings can help inform policymakers about the potential impact of changes in market parameters on market concentration, supporting the design of regulations and interventions that promote competition and market efficiency. Further research could investigate the influence of additional factors on market concentration, such as barriers to entry or firm heterogeneity, and

expand the simulation model to incorporate more complex market dynamics.

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9 Disclosure statement

The authors report there are no competing interests to declare.

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